

Advanced (Habanero)

- Q1.** What is the intersection point of the system $x + y = 2$ and $x - y = 0$?
- (A) (1, 1)
(B) (0, 0)
(C) (2, 0)
(D) (0, 2)
(E) No solution
- Q2.** Which of the following systems has no solution?
- (A) $x + y = 2$ and $x + y = 3$
(B) $x - y = 1$ and $x + y = 2$
(C) $2x + 2y = 4$ and $x + y = 2$
(D) $x + y = 2$ and $x - y = 0$
(E) $x - y = 1$ and $x + y = 1$
- Q3.** What is the solution to the system $y = 2x - 1$ and $y = -x + 2$?
- (A) (1, 1)
(B) (0, 0)
(C) (2, 3)
(D) (3, 5)
(E) (1, 1) or (0, 0)
- Q4.** The system $x + y = 4$ and $x - y = 0$ has a solution of
- (A) (2, 2)
(B) (0, 0)
(C) (4, 0)
(D) (0, 4)
(E) No solution
- Q5.** Which system has the solution (1, 3)?
- (A) $x + y = 4$ and $x - y = -2$
(B) $x + y = 2$ and $x - y = 0$
(C) $2x + y = 5$ and $x - 2y = -5$
(D) $x + 2y = 5$ and $2x - y = 1$
(E) $x - y = 1$ and $x + y = 2$
- Q6.** What is the graph of the system $y = 2x - 1$ and $y = x + 1$?
- (A) Two intersecting lines
(B) Two parallel lines
(C) A single line
(D) No lines
(E) A parabola
- Q7.** The system $x + y = 3$ and $x - y = 1$ has a solution of
- (A) (2, 1)
(B) (1, 2)
(C) (0, 0)
(D) (3, 0)
(E) (0, 3)
- Q8.** Which of the following is a solution to the system $y = x - 2$ and $y = -x + 1$?
- (A) (1, -1)
(B) (0, 0)
(C) (2, 0)
(D) (0, 2)
(E) (1, 1)

Open Ended Questions (FRQ)

Q9. Solve the system $x + 2y = 4$ and $2x - y = 1$ graphically. Show your work and box your answer.

Q10. Solve the system $y = x + 1$ and $y = 2x - 1$ graphically. Show your work and box your answer.

Chef's Tips

- Tip 1: Encourage students to use graph paper.
- Tip 2: Remind students to check their work.
- Tip 3: Encourage students to use different colors for each equation.